

| Question Number | Answer                                                                                                                                                                | Additional Guidance                                                                                                                                                                                                                                                                                                | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>3(a)</b>     | <p>An answer that includes the following:</p> <ul style="list-style-type: none"> <li>Suitable risk identified (1)</li> <li>Suitable control identified (1)</li> </ul> | <p>Any reasonable risk identified<br/>eg slips and trips, animal bites, harm from vegetation, allergies to plants, sunburn, danger from strangers, adverse weather</p> <p>Any corresponding control identified<br/>eg suitable footwear, insect repellent, suitable clothing, sunscreen, do not work alone etc</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                          | Additional Guidance | Mark       |
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| <b>3(b)(i)</b>  | <p>An answer that includes the following:</p> <p>There is no (significant) correlation between the mean length of marram grass and position on the dune (1)</p> |                     | <b>(1)</b> |

| Question Number | Answer                                                                                                                                                                                                          | Additional Guidance                                                                        | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------|
| <b>3(b)(ii)</b> | <p>An answer that includes the following:</p> <ul style="list-style-type: none"> <li>axes labelled including units, and linear scale (1)</li> <li>data plotted correctly in scatter graph format (1)</li> </ul> | <p>Mean length (of leaves) /cm and position on dune /m</p> <p>Allow one plotting error</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                | Additional Guidance                                                                                                                                                                                                                                                                          | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>3(c)(i)</b>  | <p>An answer that includes the following:</p> <ul style="list-style-type: none"> <li>• calculation of intermediate step (1)</li> <li>• calculation of <math>r_s</math> (1)</li> </ul> | <p>Example calculation</p> <p><b>Accept</b>      <math>\sum d^2 = 8</math>   <b>and</b>   <math>n=7</math></p> <p><b>OR</b>      <math>\frac{6 \sum d^2}{n(n^2 - 1)} = 0.143 / 0.14</math></p> <p><math>r_s = 0.857 / 0.86</math></p> <p>Correct answer with no working gains full marks</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                          | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                                                               | Mark       |
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| <b>3(c)(ii)</b> | <p>An explanation that includes two of the following:</p> <ul style="list-style-type: none"> <li>• correct critical value stated (1)</li> <li>• stated calculated value is greater than the stated critical value, therefore reject the null hypothesis (1)</li> <li>• there is a (significant) correlation between position on dune and (mean) length of leaves (1)</li> </ul> | <p>ie 0.786</p> <p><b>Accept</b> <math>0.857 &gt; 0.786</math>, so reject the null hypothesis for mp1 and 2</p> <p><b>Accept</b> stated calculated value is less than the stated critical value, therefore accept the null hypothesis</p> <p>Accept correct statement based on values given for bp2</p> <p><b>Accept</b> as the position on the dune increases the leaves are longer.</p> <p><b>Ignore</b> any comment on variability of data</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                             | Additional Guidance                                                                                                                                                                                                                                                                                                                                                     | Mark |
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| <b>3(d)</b>     | <p>An answer that includes three of the following:</p> <ul style="list-style-type: none"> <li>• measure more plants at each position on dune (1)</li> <li>• measure plants at more heights on the dune (1)</li> <li>• use of measuring tape instead of ruler (as some plants are more than one metre long) (1)</li> <li>• suggestion of how to avoid bias when choosing plants or leaves to measure (1)</li> </ul> | <p><b>Accept</b> repeat investigation at a different place on dune ie another transect<br/> <b>Accept</b> repeat on different dunes<br/> <b>Accept</b> measure more than 10 plants (at each position on dune)<br/> <b>Accept</b> smaller intervals or to top of dune<br/> <b>Accept</b> eg random sampling / longest leaf on each plant / other relevant suggestion</p> | (3)  |

| Question Number | Answer                                                                                                  | Additional Guidance                                                                                                        | Mark       |
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| <b>3(e)(i)</b>  | <p>An answer that includes:</p> <ul style="list-style-type: none"> <li>a suitable factor (1)</li> </ul> | <p><b>Accept</b> pH / windspeed / soil moisture / soil humus content / salinity / mineral content / water availability</p> | <b>(1)</b> |

| Question Number | Answer                                                                                                                                                                                                    | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                             | Mark       |
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| <b>3(e)(ii)</b> | <p>A description that includes the following:</p> <ul style="list-style-type: none"> <li>use of correct {equipment / method} (1)</li> <li>standardisation (1)</li> <li>location of samples (1)</li> </ul> | <p>Allow ecf from 3e(i)</p> <p><b>Accept</b> use of probe / meter if appropriate.<br/> <b>Accept</b> eg collection of soil sample, weighing, heating and reweighing</p> <p>eg<br/> sample {at known depth / meter at known height above dune surface} /<br/> sample for same time /<br/> sample at same time</p> <p>eg sample at {every / each} position on the dune / sample every 2 metres along transect</p> | <b>(3)</b> |

**(Total for question 3 = 16 marks)**

